GOVT 6019: Week 9 worksheet: OLS assumptions applied

Work in pairs. Write your answers in the boxes.

The question and the data

Children's deaths before age 5 are among the outcomes most closely tied to what governments do: vaccination campaigns, clinics, clean water, roads to reach a hospital. Do countries with more effective governments have lower child mortality? We look at 46 countries in sub-Saharan Africa in 2021, using World Bank data.

Variable	Meaning	Units
`u5mort`	Under-5 mortality rate	deaths before age 5 per 1,000 live births
`goveff`	Government Effectiveness, Worldwide Governance Indicators (WGI)	index, about −2.5 to 2.5; mean 0 and SD 1 across all countries in the world

What `goveff` measures. The World Bank defines government effectiveness as perceptions of the quality of public services, the quality of the civil service and its independence from political pressure, the quality of policy formulation and implementation, and the credibility of the government's commitment to its policies. The index combines more than 30 surveys and expert assessments. Examples of the inputs: residents' satisfaction with roads, public transportation and schools (Gallup World Poll); how well the government handles basic health services and educational needs (Afrobarometer); quality of the bureaucracy (Economist Intelligence Unit); quality of public administration and of budgetary and financial management (World Bank and African Development Bank expert ratings).

The regression in R

```
library(tidyverse)
ssa <- read_csv("wdi_ssa.csv") |> drop_na()   # the 46 countries with complete data
```

```r
fit <- lm(u5mort ~ goveff, data = ssa)
summary(fit)
```

```
Call:
lm(formula = u5mort ~ goveff, data = ssa)

Residuals:
    Min      1Q  Median      3Q     Max
-45.633 -12.907  -2.506   6.717  85.687

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   40.935      5.500   7.442 2.58e-09 ***
goveff       -31.644      5.861  -5.399 2.56e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 25.25 on 44 degrees of freedom
Multiple R-squared:  0.3985,    Adjusted R-squared:  0.3848
F-statistic: 29.15 on 1 and 44 DF,  p-value: 2.557e-06
```

$\hat{\beta}_0$ (top row) and $\hat{\beta}_1$ (bottom row)

$\widehat{\text{SE}}$: the estimated standard deviation of each estimate across samples

Estimate / $\widehat{\text{SE}}$: the t statistic for H_0: coefficient $= 0$

p-value for that t test

$\hat{\sigma}_\varepsilon = \sqrt{\text{SSR}/(n-2)}$, with $n - 2 = 44$ degrees of freedom

R^2

```r
confint(fit)
```

```
                2.5 %     97.5 %
(Intercept)  29.85044  52.02043
goveff      -43.45585 -19.83145
```

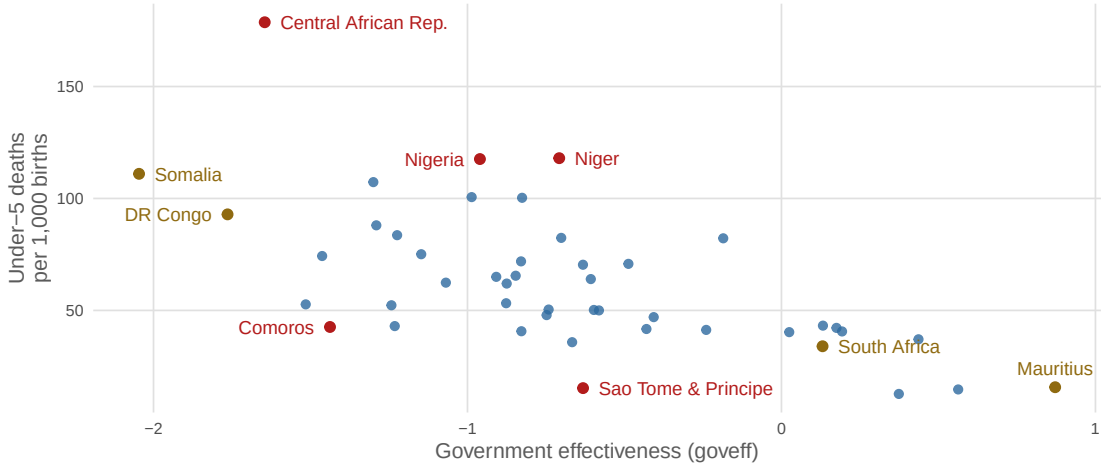
95% confidence intervals for β_0 (top row) and β_1 (bottom row)

Checking the assumptions

A1: Linearity

What it says: $\texttt{u5mort}_i = \beta_0 + \beta_1\,\texttt{goveff}_i + \varepsilon_i$. *Here:* average under-5 mortality falls along a straight line as government effectiveness rises, across the whole range of the index.

```r
ggplot(ssa, aes(x = goveff, y = u5mort)) +
  geom_point()
```



1. Do the data look roughly linear? Explain in one or two sentences.

A2: Random sampling

What it says: (X_i, Y_i), $i = 1, \ldots, 46$, is an iid sample. *Here:* the data hold every country in the region with complete data, not a random sample of countries. Treating the countries as an iid sample is an assumption about a hypothetical population of countries. (This is the 'superpopulation' view. We treat the observed units as one draw from a hypothetical process that could have produced other countries.)

2. Independence: is mortality in one country unrelated to mortality in its neighbors, once government effectiveness is known? Give an example of when this assumption might be violated.

A3: Variation in X

What it says: $\sum_i (X_i - \bar{X})^2 > 0$. *Here:* government effectiveness ranges from -2.05 (Somalia) to 0.87 (Mauritius), so A3 holds. A3 is the one assumption we can verify directly in the data, because A3 concerns only the observed values of X. For A1, A5 and A6 we look for evidence in the residuals.

A4: Zero conditional mean

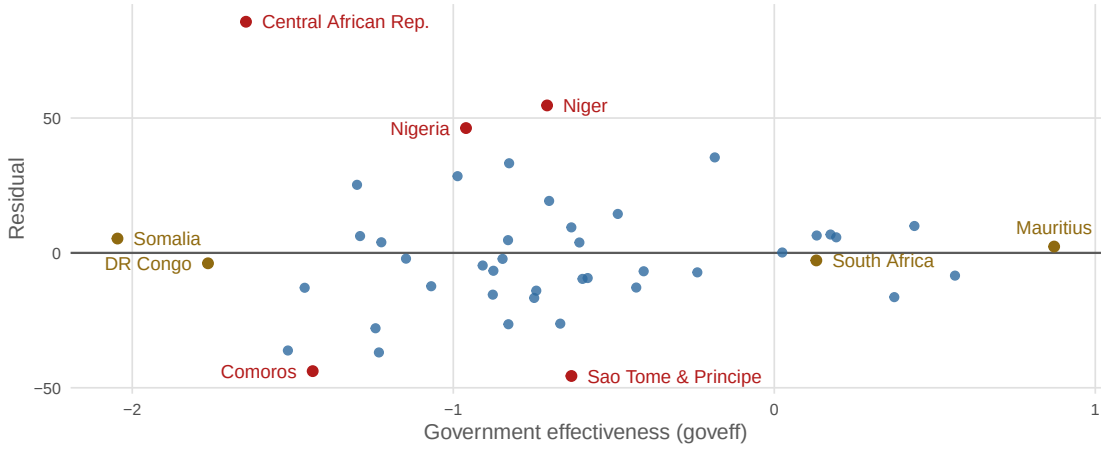
What it says: $\mathbb{E}[\varepsilon \mid \texttt{goveff} = x] = 0$ for every x. *Here:* what ε contains depends on our target. Our target is the causal effect of `goveff`: the change in child mortality if we intervened to change government effectiveness. So ε contains every other influence on child mortality. A4 says the other influences have the same average at every level of government effectiveness.

3. Name one other influence on child mortality that is likely correlated with government effectiveness. Could child mortality also affect government effectiveness? How?

A5: Homoskedasticity

What it says: $\mathbb{V}[\varepsilon \mid \texttt{goveff} = x] = \sigma_\varepsilon^2$ for every x. *Here:* the errors are equally spread out at every level of government effectiveness. We check the residuals, our stand-in for the errors.

```
ssa$resid <- resid(fit)
ggplot(ssa, aes(x = goveff, y = resid)) +
  geom_point() +
  geom_hline(yintercept = 0)
```



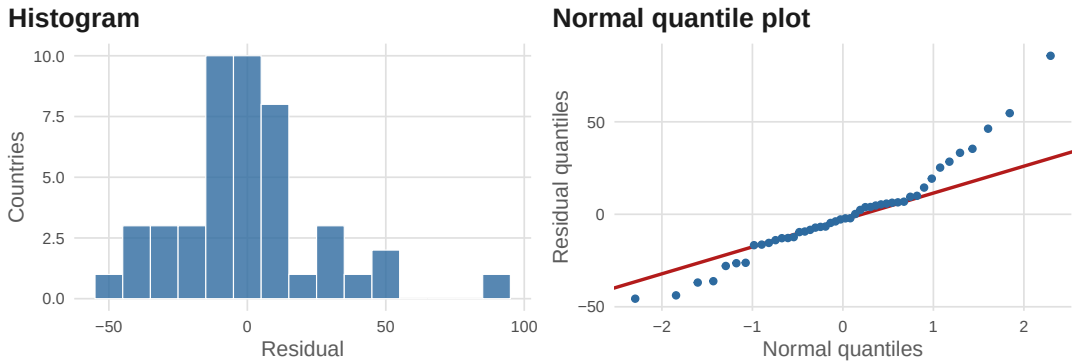
4. Does the spread of the residuals look the same at every level of government effectiveness? Where is the spread larger?

A6: Normality of the errors

What it says: the errors are normally distributed at each value of X. If n is large, we do not need A6. The sampling distribution of $\hat{\beta}_1$ is approximately normal by the central limit theorem. A common rule of thumb is $n \geq 30$. With strongly skewed errors, n must be larger. *Here:* $n = 46$ passes the rule of thumb, but the errors may be skewed, so we check whether the residuals look roughly normal.

A **normal quantile plot** sorts the residuals from smallest to largest and plots each one against the value we would expect at that rank if the residuals came from a normal distribution. If the residuals are roughly normal, the points lie near the straight line. Points that bend above the line at the right end mean a long right tail: a few residuals far larger than a normal distribution would produce.

```
ggplot(ssa, aes(x = resid)) + geom_histogram(binwidth = 10)
ggplot(ssa, aes(sample = resid)) + stat_qq() + stat_qq_line()
```



5. Do the residuals look normally distributed? Which country is the point at the top right of the quantile plot?

Interpreting the estimates

Answer as if the assumptions hold.

6. Interpret the slope in one sentence, starting "On average, . . . ".

7. Interpret the intercept. Hint: what does `goveff` $= 0$ mean?

8. Read the 95% confidence interval for the slope off the R output. What does the interval mean? Is the slope statistically significant at the 5% level?

9. Interpret R^2.

Reminder: interpreting regression

- $\hat{\beta}_0$: the predicted value of Y when $X = 0$.
- $\hat{\beta}_1$: on average, a one-unit increase in X is associated with a $\hat{\beta}_1$-unit change in Y.
- Unlike $\hat{r}$, $\hat{\beta}_1$ is in the units of Y per unit of X.
- R^2: the share of the variation in Y accounted for by X.